Rustem Islamov

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Education

PhD in Computer Science — Oct. 2023 – Pres.
University of Basel, Supervisor: Aurelien Lucchi — Basel, Switzerland

Master of Science in Applied Mathematics — Sept. 2021 – Aug. 2023
Institut Polytechnique de Paris — Palaiseau, France
Supervisor: Olivier Fercoq, Thesis supervisor: Dan Alistarh
GPA: 17.65/20, transcript of records

Bachelor of Science in Applied Mathematics and Physics — Sept. 2017 – July 2021
Moscow Institute of Physics and Technology — Dolgoprudny, Russia
Supervisor: Peter Richtárik
GPA: 4.95/5 (9.27/10), Top 3 at the department, transcript of records

Research Interests

Deep Learning (loss landscape of neural networks), Optimization (deep learning optimizers, differential privacy and robustness in distributed optimization, model quantization, computation- and memory-efficient methods)

Internships and Research Visits

Research Visit in the group of Prof. Eduard Gorbunov — Dec. 2025 – Feb. 2026
MBZUAI, Host: Eduard Gorbunov — Abu Dhabi, UAE

Internship at Distributed Algorithms and Systems Lab — Apr. 2023 – Oct. 2023
IST Austria, Supervisors: Mher Safaryan, Dan Alistarh — Klosterneuburg, Austria

Internship at Machine Learning and Optimization Lab — Apr. 2022 – Aug. 2022
EPFL, Supervisors: Hadrien Hendrikx, Martin Jaggi — Lausanne, Switzerland

Internship at Optimization and Machine Learning Lab — Mar. 2021 – Aug. 2021
KAUST, Supervisor: Peter Richtárik — Jul. 2020 – Dec. 2020
Thuwal, Saudi Arabia

Scholarships, Honors and Awards

Top Reviewer for ICML 2025, 2026, NeurIPS 2025, ICLR 2026 — 2025-2026
NeurIPS@Paris Workshop Travel Grant — Dec. 2024
French Embassy Scholarship — Sept. 2022 – May. 2023
PhD Track Excellence Scholarship — Sept. 2021 – Mar. 2022; Sept. 2022 – Mar. 2023
Increased State Academic Scholarship — Feb. 2021 – June 2021; Sept. 2020 – Jan. 2021
Abramov scholarship — Sept. 2017 – June 2020

Selected Publications (Total: 2 NeurIPS, 7 ICML, 4 ICLR, 3 AISTATS, 1 UAI, 3 TMLR)

24. K. Patel*, R. Islamov*, S. Stich, A. Lucchi, E. Gorbunov, L. Wang **What's in a Smoothness Constant? Tighter Rates for Local SGD with Bounded Second-order Heterogeneity**, arXiv preprint arXiv: 2607.14731, 2026.
23. C. Zhang*, R. Islamov*, E.M. Compagnoni, J. Pang, A. Lucchi, A. Orvieto, **Beyond a Single Explanation of the Adam-SGD Gap**, arXiv preprint arXiv: 2606.14259, 2026.
22. R. Islamov, G. Malinovsky, A. Gaponov, A. Lucchi, P. Richtárik, E. Gorbunov, **Byzantine-Robust and Differentially Private Federated Optimization under Weaker Assumptions**, In Proc. of 42nd Conference on Uncertainty in Artificial Intelligence, 2026.
21. R. Islamov, R. Machacek, A. Lucchi, A. Silveti-Falls, E. Gorbunov, V. Cevher, **On the Role of Batch Size in Stochastic Conditional Gradient Methods**, In Proc. of 43rd International Conference on Machine Learning, 2026.

20. R. Islamov, M. Crawshaw, J. Cohen, R. Gower, **Non-Euclidean Gradient Descent Operates at the Edge of Stability**, In Proc. of 43rd International Conference on Machine Learning, (**Oral**), 2026.
18. Foivos Alimisis*, R. Islamov*, A. Lucchi (*equal contribution, alphabetical order). **Why Do We Need Warm-up? A Theoretical Perspective**, In Proc. of 43rd International Conference on Machine Learning, 2025.
17. R. Islamov, N. Ajroldi, A. Orvieto, A. Lucchi. **Enhancing Optimizer Stability: Momentum Adaptation of NGN Step-size**, in Advances in Neural Information Processing Systems, 2025.
16. E. M. Compagnoni, R. Islamov, A. Orvieto, E. Gorbunov. **On the Interaction of Noise, Compression Role, and Adaptivity under -Smoothness: An SDE-based Approach**, In Proc. of 43rd International Conference on Machine Learning, 2025.
15. R. Islamov, Y. As, I. Fatkhullin. **Safe-EF: Error Feedback for Non-smooth Constrained Optimization**, In Proc. of 42nd International Conference on Machine Learning, 2025.
14. R. Islamov, S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov. **Double Momentum and Error Feedback for Clipping with Fast Rates and Differential Privacy**, arXiv preprint arXiv: 2502.11682, 2025.
13. E. M. Compagnoni, R. Islamov, F. N. Proske, A. Lucchi. **Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs**, in Proc. of the 28th International Conference on Artificial Intelligence and Statistics, (**Oral**), 2025.
12. E. M. Compagnoni, T. Liu, R. Islamov, F. N. Proske, A. Orvieto, A. Lucchi. **Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise**, in Proc. of the 13th International Conference on Learning Representations, 2024.
11. R. Islamov, N. Ajroldi, A. Orvieto, A. Lucchi. **Loss Landscape Characterization of Neural Networks without Over-Parametrization**, in Advances in Neural Information Processing Systems, 2024.
10. R. Islamov*, Y. Gao*, S. Stich(*equal contribution). **Towards Faster Decentralized Stochastic Optimization with Communication Compression**, in Proc. of the 13th International Conference on Learning Representations, 2024.
9. Y. Gao*, R. Islamov*, S. Stich (*equal contribution). **EControl: Fast Distributed Optimization with Compression and Error Control**, in Proc. of the 12th International Conference on Learning Representations, 2023.
8. R. Islamov, M. Safaryan, D. Alistarh. **AsGrad: A Sharp Unified Analysis of Asynchronous-SGD Algorithms**, in Proc. of the 27th International Conference on Artificial Intelligence and Statistics, 2023.
7. K. Mishchenko, R. Islamov, E. Gorbunov, S. Horváth. **Partially Personalized Federated Learning: Breaking the Curse of Data Heterogeneity**, Transactions on Machine Learning Research, 2023.
6. S. Khirirat, E. Gorbunov, S. Horváth, R. Islamov, F. Karray, P. Richtárik. **Clip21: Error Feedback for Gradient Clipping**, arXiv preprint arXiv: 2305.18929, 2023.
5. M. Makarenko, E. Gasanov, R. Islamov, A. Sadiev, P. Richtárik. **Adaptive Compression for Communication-Efficient Distributed Training**, Transactions on Machine Learning Research, 2022.
4. R. Islamov, X. Qian, S. Hanzely, M. Safaryan, P. Richtárik. **Distributed Newton-Type Methods with Communication Compression and Bernoulli Aggregation**, Transactions on Machine Learning Research, 2022.
3. X. Qian, R. Islamov, M. Safaryan, P. Richtárik. **Basis Matters: Better Communication-Efficient Second Order Methods for Federated Learning**, in Proc. of the 25th International Conference on Artificial Intelligence and Statistics, 2022.
2. M. Safaryan, R. Islamov, X. Qian, P. Richtárik. **FedNL: Making Newton-Type Methods Applicable to Federated Learning**, In Proc. of 39th International Conference on Machine Learning, 2022.
1. R. Islamov, X. Qian, P. Richtárik. **Distributed Second Order Methods with Fast Rates and Compressed Communication**, In Proc. of 38th International Conference on Machine Learning, 2021.

Reviewing and Teaching

Teaching Assistant for Continuous Optimization course	Spring Semesters 2024-2026
Reviewer for Transactions on Machine Learning Research (TMLR)	2023-2026
Reviewer for Conference on Neural Information Processing Systems	2024-2025
Reviewer for International Conference on Machine Learning (ICML)	2024-2026
Reviewer for Journal on Machine Learning Research (JMLR)	2024-2025
Reviewer for Journal of Parallel and Distributed Computing (JPDC)	2024
Reviewer for Journal of Optimization Theory and Applications (JOTA)	2023
Reviewer for Artificial Intelligence and Statistics Conference (AISTATS)	2023-2026
Reviewer for Neural Information Processing Systems (NeurIPS)	2024-2026
Reviewer for International Conference on Learning Representations (ICLR)	2024-2026

Talks and Posters

Talk at Swiss Optimization Symposium, Links: paper	23 August, 2026
Lightning Talk at Protocol Learning: Decentralized Collaborative Learning at Scale (ICML 2026 Workshop) Links: paper	10 July, 2026
Oral Presentation at ICML 2026 Conference, Links: paper	9 July, 2026
Invited Talk at SIAM OP26 Conference, Links: paper	2-5 June, 2026
Invited Talk at EUROPT Conference, Links: paper	29 June-2 July, 2025
Invited Talk at Rising Stars in AI Symposium at KAUST workshop, Links: paper, slides	21 February, 2024
Talk at NTDS workshop, Links: paper	24 October, 2023
Invited Talk at CISPA, Links: paper	16 March, 2023
Talk at ETH AI Center Symposium for PhD fellows, Links: paper	9-10 February, 2023
Prerecorded Talk at KAUST Conference on Artificial Intelligence, Links: video, paper	28 April, 2021

Technical skills

Programming Languages: Python (NumPy, Matplotlib, PyTorch, Pandas), C++, LaTeX
Mathematics: Calculus, Linear Algebra, Probability Theory, Convex Analysis, Deep Learning

Languages

Russian: Native
English: Advanced (C1)
French: Elementary (A2)

Hobbies and Interests

Football, former member of student football team
Travelling, hiking, photo shooting